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Causal Sufficient Dimension Reduction for Multiple Continuous Exposures

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Motivation

Environmental mixture studies ask how health outcomes change under exposure to multiple pollutants. The National Institute of Environmental Health Sciences (NIEHS) is interested in understanding the *joint effect of exposures* instead of treating pollutants in isolation.

A common strategy follows a linear additive confounding model. Suppose Y is an outcome, X are the exposures, and C are confounders:

$$Y = h(X) + \gamma^\top C + \varepsilon, \quad X \in \mathbb{R}^p.$$

These methods differ in how they handle the true exposure dimension d and complexity of $h(\cdot)$, creating tradeoffs between the flexibility, interpretability, and causal validity of the estimated relationship between exposure and outcome. They include full exposure-response surface (ERS), single-index, and dimension reduction methods.

Method class	Flexible	Interpretable	Supervised	Causal
Full ERS ($d = p$)	✓	×	✓	×
Single-index ($d = 1$)	×	✓	✓	×
Dimension reduction ($1 \leq d \leq p$)	✓	✓	×	×
CSDR (ours)	✓	✓	✓	✓

GOAL: Use sufficient dimension reduction (SDR) to learn a low-rank summary of the original exposure X that preserves the ERS, simultaneously achieving flexibility, interpretability and valid causal inference.

Background

We propose **causal sufficient dimension reduction (CSDR)** to address these challenges. Let Y^x be the potential outcome under exposure $X = x$. The causal exposure-response surface (ERS) is

$$\mu(x) = \mathbb{E}(Y^x), \quad \mu: \mathbb{R}^p \rightarrow \mathbb{R}.$$

CSDR assumes the ERS depends on X only through a low-dimensional index,

$$\mu(x) = g(\beta^\top x), \quad \beta \in \mathbb{R}^{p \times d}, \quad d < p, \quad (1)$$

for some smooth function $g(\cdot)$. The estimand is not a particular basis β , but the causal central mean subspace (CCMS) or $\mathcal{S}_{\mathbb{E}(Y^x)}$, the intersection of all $\text{span}(\beta)$ for matrices β that satisfy (1).

Nabi et al. (2022) proposed estimating $\mathcal{S}_{\mathbb{E}(Y^x)}$ with a locally efficient score based on marginal structural models [1]. It applies inverse propensity weighting and g -comp to the noncausal score for β :

Nuisance functions:

$$\begin{aligned} \mu_\beta(z) &:= \mathbb{E}(Y^z) && \text{(causal ERS),} \\ \nu_\beta(z) &:= \mathbb{E}\{\alpha(X)^z\} && \text{(causal exposure regression),} \\ \pi(X|C) &:= f_{X|C}(X|C) && \text{(generalized propensity score),} \\ m(X, C) &:= \mathbb{E}(Y|X, C) && \text{(outcome regression).} \end{aligned}$$

Core noncausal score component:

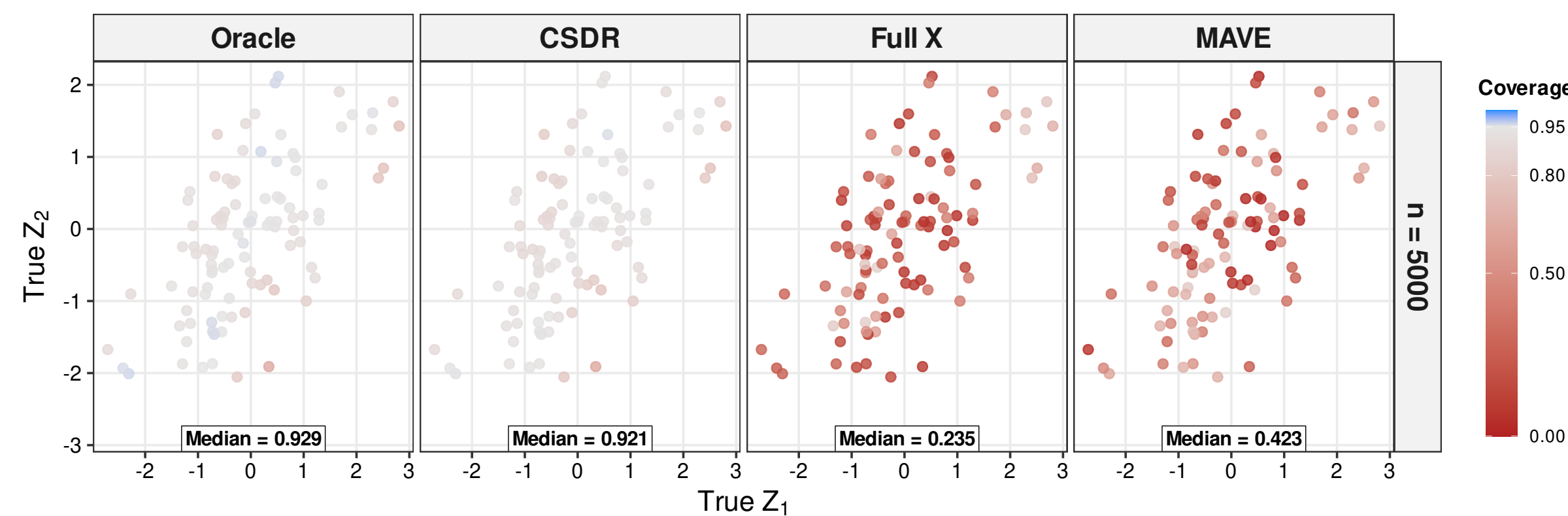
$$U(\beta) = \{Y - \mu_\beta(Z)\}\{\alpha(X) - \nu_\beta(Z)\}.$$

Locally efficient causal score:

$$\tilde{U}(\beta) = \frac{\pi^*(X)}{\pi(X|C)} \left\{ U(\beta) - \mathbb{E}[U(\beta) | X, C] \right\} + \mathbb{E}^*[\mathbb{E}[U(\beta) | X, C] | C].$$

The score is challenging to implement due to 1) unclear choice of $\alpha(\cdot)$ and $\pi^*(\cdot)$ and 2) estimation of β -dependent nuisances. We propose an alternative two-stage approach.

Simulation and Real Data Analysis Results



Simulations ($p = 10, d_0 = 2$) compare CSDR with the Oracle (true β is known), association-based dimension reduction, and causal inference using the full X vector. CSDR recovers $\mathcal{S}_{\mathbb{E}(Y^x)}$ and selects the correct dimension with performance similar to the Oracle. Causal dimension reduction improves 95% coverage compared to causal inference using X without dimension reduction.

Causal Sufficient Dimension Reduction

Observed data and causal target
(Y_i, X_i, C_i) with $Y_i \in \mathbb{R}, X_i \in \mathbb{R}^p, C_i \in \mathbb{R}^q$
Target: A basis matrix β of $\mathcal{S}_{\mathbb{E}(Y^x)}$ where $\mu(x) = \mathbb{E}(Y^x) = g(\beta^\top x), d < p$

Stage 1: Construct ($\tilde{Y}, \tilde{X}$) so that $\mathbb{E}(\tilde{Y} | \tilde{X} = x) = \mu(x)$.

Options include the regression adjustment (RA) estimator, the doubly robust estimator (DR) (not shown), pseudo-outcome (PO), and residualized pair (RP).

Variant	$\tilde{Y}_i$	$\tilde{X}_i$
RA	$\mu(X_i) = n^{-1} \sum_{i=1}^n \hat{m}(X_i, C_i)$	X_i
PO	$\frac{Y_i - \hat{m}(X_i, C_i)}{\hat{\pi}(X_i C_i)} \left(n^{-1} \sum_{i=1}^n \hat{\pi}(X_i C_i) \right) + n^{-1} \sum_{i=1}^n \hat{m}(X_i, C_i)$	X_i
RP	$Y_i - \hat{\mathbb{E}}(Y C_i)$	$X_i - \hat{\mathbb{E}}(X C_i)$

Stage 2: apply association-based SDR to ($\tilde{Y}, \tilde{X}$)

A local linear approximation to $g(\beta^\top x)$ recovers β where w_{ij} are kernel weights. Minimum average variance estimation (MAVE) efficiently solves for β [2].

$$\hat{\beta}_d = \arg \min_{\substack{\beta^\top \beta = I_d \\ a_1, \dots, a_n, b_1, \dots, b_n}} \sum_{j=1}^n \sum_{i=1}^n [\tilde{Y}_i - \{a_j + b_j^\top \beta^\top (\tilde{X}_i - \tilde{X}_j)\}]^2 w_{ij}^{(d)}.$$

We also estimate the true dimension d using a Nadaraya-Watson estimator.

$$\hat{d} = \arg \min_{1 \leq d \leq p} CV(d), \quad CV(d) = \frac{1}{n} \sum_{j=1}^n \{\tilde{Y}_j - \hat{\mathbb{E}}(\tilde{Y}_j | \hat{\beta}_d^\top \tilde{X}_j)\}^2.$$

Subspace Importance Score (SIS)

We analyze each original variable's causal importance to $\mathcal{S}_{\mathbb{E}(Y^x)}$ using the SIS defined as $\lambda_j := [\hat{P}]_{jj}$ where $\hat{P} := \hat{\beta}(\hat{\beta}^\top \hat{\beta})^{-1} \hat{\beta}^\top$:

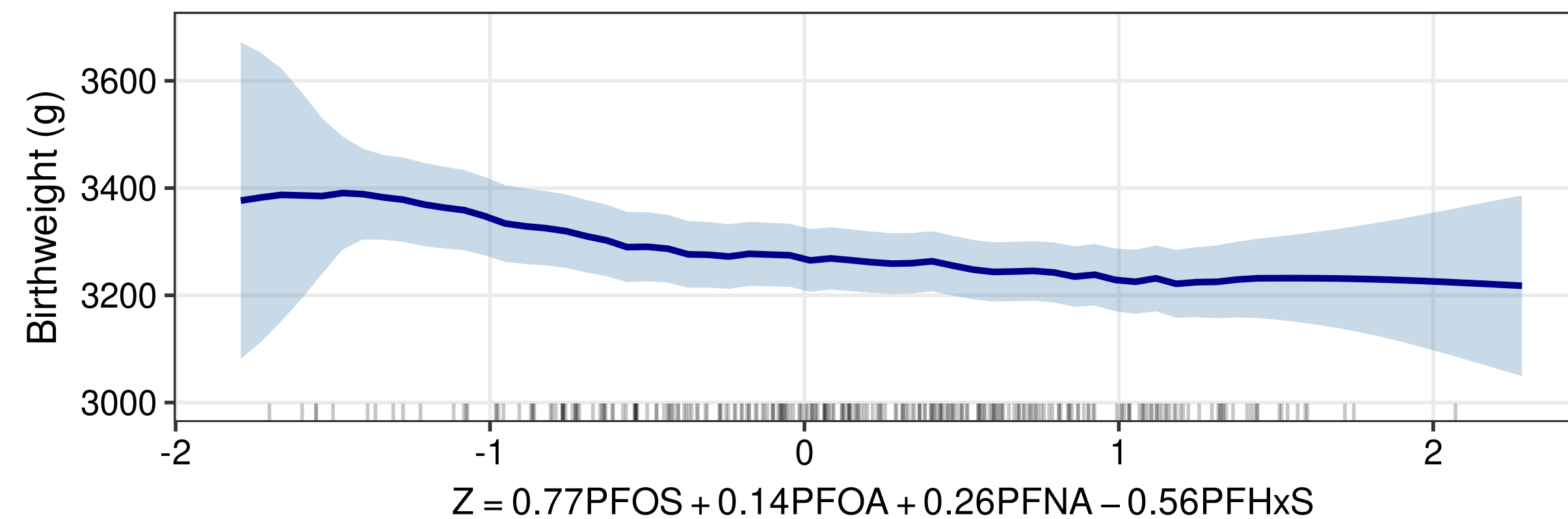
- **Basis-invariant:** depends only on $\text{span}(\hat{\beta})$, not the particular basis β .
- **Bounded:** $0 \leq \lambda_j \leq 1$ by property of projection matrix.
- **Additive:** $\sum_{j=1}^p \lambda_j = \hat{d}$, so total importance is an additive decomposition of d .
- **Interpretable:** larger λ_j means X_j contributes more.

Post-reduction causal ERS estimation

Define the reduced exposure $\hat{Z} := \hat{\beta}^\top X$ and use a continuous exposure causal estimator with Z replacing X [3]. The smoothing dimension is now d , not p .

$$\hat{\mu}_{\hat{\beta}, DR}(z) = \frac{1}{n} \sum_{i=1}^n \left[\frac{K_{h_Z}(\hat{Z}_i - z)}{h_Z^d \hat{\pi}_{\hat{\beta}}(\hat{Z}_i | C_i)} \{Y_i - \hat{m}_{\hat{\beta}}(z, C_i)\} + \hat{m}_{\hat{\beta}}(z, C_i) \right].$$

Pointwise 95% CIs are generated using the sample variance of the score. Uniform 95% CIs can be formed using the multiplier bootstrap.



We applied CSDR to a cohort of 305 African-American mothers to estimate the joint effect of four PFAS exposures on infant birthweight. We estimated a 1D PFAS mixture with $\hat{Z} = 0.77 \text{ PFOS} + 0.14 \text{ PFOA} + 0.26 \text{ PFNA} - 0.56 \text{ PFHxS}$. Birthweight decreased with increasing PFAS exposure, and the SIS values $\lambda = (0.60, 0.02, 0.07, 0.31)$ identified PFOS as the dominant contributor.

Theory and Assumptions

Assumptions:

► **(Causal identification - standard)** (i) consistency $Y = Y^X$, (ii) conditional ignorability $Y^x \perp X | C$, and (iii) strong positivity $\pi(x|c) \geq \epsilon > 0$ on the support. These ensure the causal ERS $\mu(x) = \mathbb{E}(Y^x)$ is identifiable.

► **(Causal identification - smoothness):** (i) $m(x, c)$ and $\pi(x|c)$ have bounded, continuous third derivatives with respect to x . (ii) $\text{Var}(Y|X, C)$ and its derivatives w.r.t. X are bounded uniformly. These are needed for identification of continuous exposures.

► **(Dimension reduction structure):** (i) the ERS can be written as $g(\beta^\top x)$ for some $d < p$ (ii) the reduced exposure $Z = \beta^\top X$ is well-defined and inherits positivity and conditional ignorability.

► **(Smoothness for SDR):** (i) marginal density of $f_X(\cdot)$ has bounded derivatives up to order 4 and compact support, (ii) $g(\cdot)$ has bounded continuous derivatives up to order 3, (iii) kernels are regular, and (iv) the first-stage estimator of $\tilde{Y}$ set to $\mu(X)$ satisfies a uniform rate $R_n = \sup_i |\hat{\mu}(X_i) - \mu(X_i)| = o(h^2)$.

Identification. The causal ERS is identified by:

$$\mu(x) = \mathbb{E}\{m(x, C)\} = \lim_{h \rightarrow 0} \mathbb{E} \left[\frac{1}{h^p} \frac{K_h(X - x)}{\pi(X|C)} Y \right].$$

For a reduced exposure Z , the same representation holds on the lower-dimensional support:

$$\mu_\beta(z) = \mathbb{E}\{m_\beta(z, C)\} = \lim_{h \rightarrow 0} \mathbb{E} \left[\frac{1}{h^d} \frac{K_h(Z - z)}{\pi_\beta(Z|C)} Y \right].$$

Theorem 1 (Convergence rate of $\hat{\beta}$). Let β span the true CCMS and $\hat{\beta}$ be the csMAVE estimate based on the ERS pair. If δ_n is $\{\log n / (nh^p)\}^{1/2}$, then

$$\| (I_p - \hat{\beta} \hat{\beta}^\top) \beta \| = O_p(h^3 + h \delta_n + h^{-1} R_n).$$

Compare to the original MAVE convergence rate which is

$$O_p(h^3 + h \delta_n + h^{-1} \delta_n^2).$$

The first two terms are standard MAVE error; the $h^{-1} R_n$ term replaces the old term and captures the impact of estimation from the first stage.

Theorem 2 (Consistency of $\hat{d}$). If the first-stage ERS estimator satisfies $R_n = o_p(h^2)$, then the MAVE dimension selector remains consistent:

$$\Pr(\hat{d} = d_0) \rightarrow 1.$$

Key Points

► **Computational efficiency:** we have developed a two-stage estimator for $\mathcal{S}_{\mathbb{E}(Y^x)}$ that reduces the nuisance estimation and numerical search burden of the previous MSM approach.

► **Recovers the causal subspace:** The method consistently estimates the CCMS and its dimension d with error dependent on how well $(\tilde{Y}, \tilde{X})$ are estimated.

► **Modular and flexible:** any valid first-stage causal estimator (RA, DR, PO, RP) can be combined with regular association-based SDR methods like MAVE, enabling robustness and adaptability.

► **Dimension reduction improves inference:** projecting to $Z = \beta^\top X$ reduces the smoothing dimension from p to d , improving nuisance estimation, efficiency and inference.

► **Variable importance:** Basis-invariant SIS λ_j gives a transparent and interpretable variable importance measure.

References

[1] Nabi, R., McNutt, T., and Shpitser, I. (2022). Semiparametric causal sufficient dimension reduction of multidimensional treatments. Proceedings of the 38th Conference on UAI.

[2] Xia, Y., Tong, H., Li, W. K., and Zhu, L.-X. (2002). An Adaptive Estimation of Dimension Reduction Space. Journal of the Royal Statistical Society B (JRSS-B).

[3] Colangelo, K. and Lee, Y.-Y. (2025). Double Debiased Machine Learning Nonparametric Inference with Continuous Treatments. Journal of Business & Economic Statistics.

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